

Seeing Like an Organization (#269965)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has 3 authors.

Pre-registered on:

2026/01/26 07:18 (PT)

1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

We test whether shared cognitive schemas can both enable and hinder adaptation under environmental change. Participants are randomized into small groups and assigned to one of two conditions (Schema vs. Non-Schema). Each group completes four rounds of a coordination task. The first two rounds are condition-specific: in the Schema condition, rounds 1-2 share a consistent feature structure that encourages schema formation; in the Non-Schema condition, rounds 1-2 lack such structure and discourage schema formation. Rounds 3-4 are identical across conditions. Round 3 introduces an environmental change but remains solvable using schemas developed for rounds 1-2 of the Schema condition ("schema-conserving" change). Round 4 introduces a change that may appear solvable using that schema, but is not.

H1: Groups in the Schema condition will outperform groups in the Non-Schema condition under schema-conserving environmental change (round 3) by completing the labeling task in less time.

H2a: The performance advantage of groups in the Schema condition over groups in the Non-Schema condition will attenuate under schema-destroying environmental change (round 4).

H2b: Groups in the Non-Schema condition will outperform groups in the Schema condition under schema-destroying environmental change by correctly matching on more correct and unique names during the recall stage. (round 4)

3) Describe the key dependent variable(s) specifying how they will be measured.

Each round of the coordination task will be composed of two stages. First, during the Discussion Stage, participants will engage in a discussion and be asked to name a set of 8 images such that each image receives a unique name. Second, during the Recall Stage, participants will be asked to recall the names given to each image, one by one ("recall stage"). Outcomes will be analyzed at the group level. For each round, we compute: (1) Accuracy - the number of images (out of 8) for which all group members recalled the same unique name in the Recall Stage (i.e., no name repetition); (2) Labeling Time - the time spent in the Discussion Stage (measured as the maximum time taken by either partner in each group)

4) How many and which conditions will participants be assigned to?

The participants will be assigned to one of two conditions: (1) Schema - In the first two rounds of the game, participants will see sets of images that can be differentiated using the same set of salient features; (2) Non-schema - In the first two rounds of the game, participants will see sets of images that cannot be differentiated using the same set of salient features

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

To test H1, we will compare Round 3 labeling time between Schema and Non-Schema conditions using a one-tailed Welch's t-test at $\alpha = 0.05$, testing whether Schema < Non-Schema. We will report group means, the mean difference (Schema - Non-Schema) with a 95% confidence interval, and Cohen's d as an effect size.

To test H2a/H2b, we will compare Round 4 accuracy between Schema and Non-Schema conditions using a one-tailed Mann-Whitney U test at $\alpha = 0.05$, testing whether Schema < Non-Schema. We will use a permutation-based one-sided p-value with standard tie handling. We will report group medians, the Hodges-Lehmann estimate of the location shift (median of all pairwise differences, Schema - Non-Schema) with a 95% confidence interval, and Cliff's delta as a nonparametric effect size.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We will exclude groups where: (1) One or more participants are reported as not participating in the Discussion Stage, and don't check in within 60 seconds, or are reported as not participating more than once; (2) One or more participant submits fewer than 13 recall submissions during the Recall Stages (out of 38); or, (3) Video recordings reveal cheating. E.g.,: recording names and images during the Discussion Stage of the round, or communicating during the Recall Stage of the round.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

Our final sample size for collected data will be approx. $N = 128$ participants after excluding outliers (see 6 above) (approx. 64 per condition).

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

(1) Pilot data were collected prior to pre-registration for QA purposes. It will not be used in the analysis; (2) At the end of the experiment, participants will be asked to complete a demographic survey including questions on Age, Gender, Education, and Race. These data may be used for ex post exploratory analysis. They will also be asked to answer open-ended questions explaining their choices in stage 2. This data may be used in future exploratory analysis; (3) Participants will be able to communicate with one another via video chat & text boxes during the first stage. This video & textual data, along with the names participants assigned to images, will also be collected for potential future exploratory analysis.